Machine Learning Course Project Guidelines

Spring 2026

Overview

The course project is a semester-long individual effort where you will apply machine learning techniques to solve a real-world problem. You will complete this project throughout the semester and submit a final assignment summarizing your work.

Project Requirement: You must apply **at least 3 methods** learned in class to your chosen dataset. Methods covered in this course include: k-Nearest Neighbors (kNN), Linear Regression, Support Vector Machine (SVM), K-means Clustering, Decision Trees, Hierarchical Clustering, DB-SCAN, Logistic Regression, and Neural Networks.

Project Related Questions

You will answer the following questions on the D2L page to plan your project. These questions will help you define your project scope and ensure you have a viable path to completion.

1. Project Title and Abstract

Provide a descriptive title for your project and a brief abstract summarizing what you plan to accomplish.

2. Problem Statement & Motivation

- What specific problem are you trying to solve?
- Why is this problem important? (e.g., societal impact, business value, scientific discovery)
- Is this a classification, regression, clustering, or generation task?

3. Dataset Description

Crucial Step: Projects often fail due to poor data. Be specific.
- **Source:** Where is the data coming from? (e.g., Kaggle, UCI Repository, scraped data, API).
- **Size:** How many samples (rows) and features (columns) do you expect?
- **Labeling:** For supervised learning, do you have ground-truth labels?
- **Preprocessing:** What cleaning or feature engineering might be required?

4. Proposed Methodology

You must apply at least 3 methods from the course. Consider including both clustering and supervised methods.

- **Baseline:** What simple model will you use to establish a baseline? (e.g., "I will use Logistic Regression as a baseline.")
- **Additional Methods:** What other methods do you plan to apply? (e.g., "I plan to implement kNN, Decision Trees, and K-means clustering to compare against the baseline.")
- **Justification:** Why are these methods appropriate for your data?

5. Evaluation Plan

You must plan how to evaluate and compare your models. Your final assignment will require quantitative results, visualizations, and analysis.

- **Metrics:** Which metrics will you use? Choose appropriate metrics for your task:
  - *Classification:* Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix
  - *Regression:* MSE, RMSE, MAE, R^2 Score
  - *Clustering:* Silhouette Score, Inertia, Davies-Bouldin Index
- **Validation:** How will you split your data? (e.g., 80/20 split, k-fold cross-validation)
- **Comparison:** How will you compare the 3+ methods you implement?

6. Expected Results and Deliverables

Your final assignment must include:

- **Quantitative Results:** Tables comparing all methods using your chosen metrics
- **Visualizations:** Confusion matrices, ROC curves, learning curves, or clustering plots
- **Analysis:** Interpretation of results, discussion of which methods performed best and why

Recommended Datasets

To ensure your project is feasible and allows comparison of multiple ML methods, consider using one of these curated datasets:

Heart Disease Dataset

- **Source:** https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset
- **Size:** 303 instances, 14 features
- **Task:** Binary classification (heart disease presence)
- **Applicable Methods:** Logistic Regression, SVM, Decision Trees, kNN, Neural Networks

Titanic Dataset

- **Source:** https://www.kaggle.com/c/titanic/data
- **Size:** Approximately 900 samples
- **Task:** Binary classification (survival prediction)
- **Applicable Methods:** Logistic Regression, Decision Trees, kNN, SVM
- **Note:** Good practice for feature engineering with missing data

Adult Income Dataset (Census Income)

- **Source:** https://archive.ics.uci.edu/ml/datasets/adult
- **Size:** 48,842 instances with mix of categorical and continuous features
- **Task:** Binary classification (income ¿50K or ¡=50K)
- **Applicable Methods:** Logistic Regression, Decision Trees, SVM, kNN, Neural Networks
- **Note:** Requires preprocessing of categorical variables

You may also propose your own dataset, but it must allow application of at least 3 methods from the course.